

Lecture notes for Amari model: equilibrium solutions and their stability

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1 Equilibrium solutions

Amari model is given as follows:

$$\mu \frac{\partial u}{\partial t}(x, t) = -u(x, t) + \int_{-\infty}^{\infty} s(u(y, t)) \omega(x - y) dy + h + P(x, t). \quad (1)$$

Here $x \in \mathbb{R}$ and $t \in [0, \infty)$. We refer to the slides for the description of the terms appearing in this system. We will focus on the equilibrium case ($\frac{\partial u(x, t)}{\partial t} = 0$ for all $t > t_0 > 0$ where t_0 is sufficiently large¹) in the absence of time-dependent external stimulus (i.e., $P(x, t) = 0$ for all $x \in \mathbb{R}$ and $t \in [0, \infty)$).

In this equilibrium case, (1) becomes

$$u(x, t) = \int_{-\infty}^{\infty} s(u(y, t)) \omega(x - y) dy + h, \quad (2)$$

where h is a constant. In this setting, we consider the neurons at position $x \in \mathbb{R}$ at time $t \in [0, \infty)$ as **excited** if

$$u(x, t) > 0.$$

Otherwise, they are inhibited or in resting state:

$$u(x, t) \leq 0.$$

We will define the region where the neurons are excited at time t as

$$R_t(u) := \{x \in \mathbb{R} : u(x, t) > 0\}.$$

Let us recall that the transfer function s is defined as the Heaviside step function:

$$s(u) := \begin{cases} 0, & u \leq 0 \\ 1, & u > 0 \end{cases}.$$

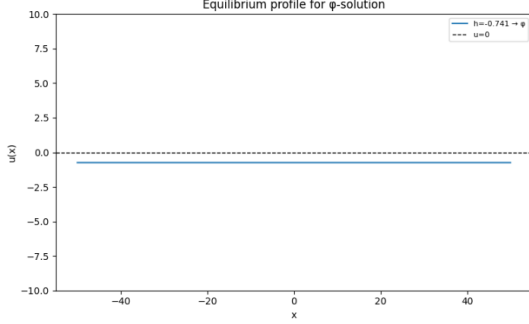
Since $s(u) = 1$ only in the excited regions and $= 0$ elsewhere, we can write (2) by restricting the integral to $R_t(u)$:

$$u(x, t) = \int_{R_t(u)} s(u(y, t)) \omega(x - y) dy + h. \quad (3)$$

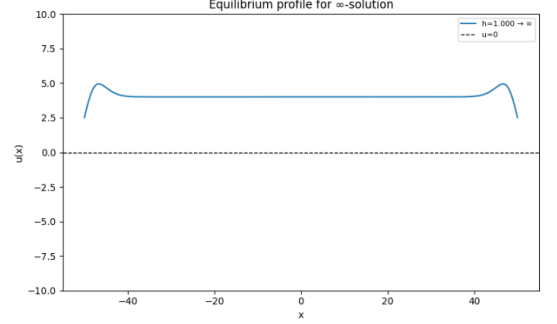
There are three categories of equilibrium solutions at time t :

¹The system reaches to an equilibrium possibly after a certain instant which we denote by t_0 . In the simulations, final time of the simulation should be chosen larger than t_0 to observe the equilibrium state of the system. Here we will continue to use the notation with time variable. However, in the literature, since in the equilibrium case the solution does not change in time, a simplified notation is used where the time variable is dropped: $u(x, t) = u(x)$.

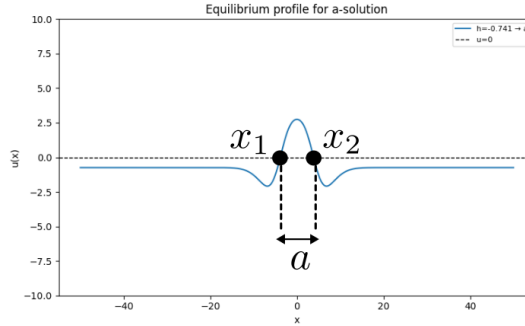
- $\emptyset$ – **solution.** $R_t(u) = \emptyset : u(x, t) \leq 0$ for all $x \in \mathbb{R}$ (Fig 1a).
- ∞ – **solution.** $R_t(u) = (-\infty, \infty) : u(x, t) > 0$ for all $x \in \mathbb{R}$ (Fig 1b).
- a – **solution.** $R_t(u) = (x_1, x_2) : u(x, t) > 0$ for all $x \in (x_1, x_2)$ where $x_1 < x_2 \in \mathbb{R}$ with $x_2 - x_1 = a$, and $u(x, t) \leq 0$ elsewhere (Fig 1c).



(a) Example of $\emptyset$ – solution



(b) Example of ∞ – solution



(c) Example of a – solution. This type of solution refers to a local excitation of length a .

Figure 1: Examples of equilibrium solution types.

For a – solution, we can assume that $R_t(u) = (0, a)$ without losing the generality since the system does not depend on the specific positions of the boundaries x_1 and x_2 corresponding to the excited region, but it rather depends on the length $a = x_2 - x_1$ of the excited region in the equilibrium case. **From now on, as long as otherwise is not stated explicitly, it should be tacitly understood that $R_t(u) = (0, a)$ for an a – solution, where $a > 0$.**

The structure of the neural field is characterized by the connectivity kernel ω . In the original work of Amari [1], ω was chosen as a Mexican-hat function² (Fig 2):

$$\omega(x) := \frac{2}{\sqrt{3}\sigma\pi^{1/4}} \left(1 - \frac{x^2}{\sigma^2}\right) e^{-\frac{x^2}{2\sigma^2}},$$

where $\sigma > 0$ is the spatial scale. The scale determines the range of excitatory and inhibitory interactions in the spatial domain.

²Mexican-hat can be approximated by difference of Gaussian functions, i.e., $\beta_e e^{-\frac{x^2}{2\sigma_e^2}} - \beta_i e^{-\frac{x^2}{2\sigma_i^2}}$. Here $\beta_e > 0$ and $\beta_i > 0$ determine the weights of excitatory and inhibitory interactions, respectively. The spatial scale $\sigma_e > 0$ and $\sigma_i > 0$ determines the ranges of excitatory and inhibitory interactions, respectively.

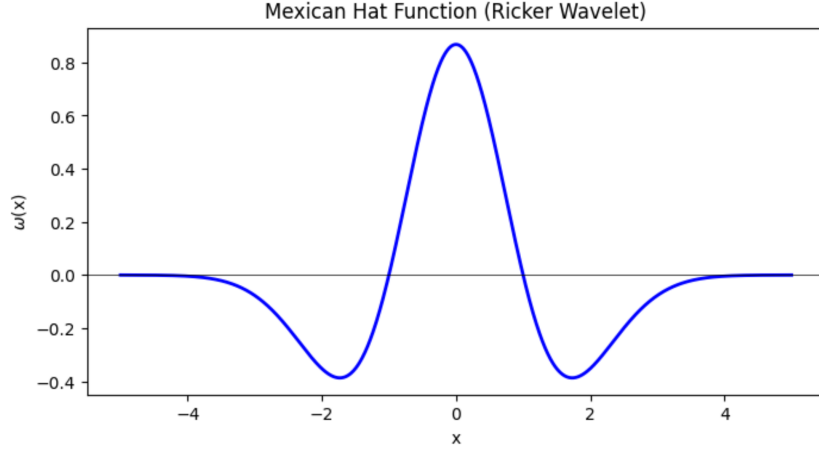


Figure 2: Plot of Mexican-hat function. It introduces a connectivity structure based on short range excitation and long range inhibition.

We define the integral of ω in the right half-plane as follows:

$$W(x) := \int_0^x \omega(y) dy.$$

Remark 1.

- $W(0) = 0$
- $W(x) = -W(-x)$ (antisymmetry)
- $W(x)$ has only one peak.

◇

We define two important quantities (Fig 3):

$$W_m := \max_{x>0} W(x), \quad W_\infty := \lim_{x \rightarrow \infty} W(x).$$

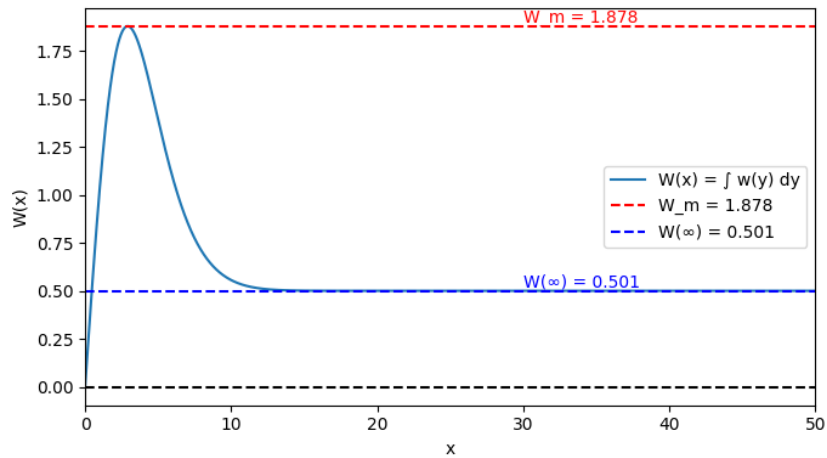


Figure 3: Plot of $W(x)$ obtained with an arbitrary Mexican-hat function.

The conditions for the existence of the equilibrium solutions in the absence of external stimulus are given by the following theorem in [1]:

Theorem 1.

In the absence of time-dependent external stimulus, i.e., $P(x, t) = 0$ for all $x \in \mathbb{R}$, $t \in [0, \infty)$, the following holds:

1. There exists an $\emptyset$ – solution $\iff h \leq 0$.
2. There exists an ∞ – solution $\iff 2W_\infty > -h$
3. There exists an a – solution $\iff h < 0$ and $a > 0$ satisfies $W(a) + h = 0$.

◇

Proof. We start with the first item. If there exists an $\emptyset$ – solution, then (3) becomes for the $\emptyset$ – solution as follows:

$$u(x, t) = h.$$

This requires that $h \leq 0$ for $u(x, t)$ to be $\emptyset$ – solution. So, if there exists an $\emptyset$ – solution, then $h \leq 0$ must hold. On the other side, if $h \leq 0$, then $u(x, t) = h$ is indeed an $\emptyset$ – solution to (3), providing that there exists an $\emptyset$ – solution.

We follow a similar reasoning in the second item. If there exists an ∞ – solution, then we can write (3) for this solution by using the antisymmetry of $W(x)$ as

$$u(x, t) = 2W_\infty + h.$$

Since $u(x, t)$ is an ∞ – solution, the following hold:

$$2W_\infty + h > 0,$$

i.e., $2W_\infty > -h$. On the other way around, if $2W_\infty > -h$, then $u(x, t) = 2W_\infty + h$ is indeed an ∞ – solution to (3).

In the third item, if there exists a – solution with $R_t(u) = (0, a)$, then (3) becomes

$$u(x, t) = \int_0^a s(u(y, t)) \omega(x - y) dy + h = W(x) - W(x - a) + h, \quad (4)$$

where we used a change of variables³ in the integral. Being an a – solution, once it reaches to equilibrium, u satisfies (e.g., as in Fig 1c)

$$u(0, t) = u(a, t) = 0, \quad (5)$$

This provides $u(a, t) = W(a) - W(0) + h = W(a) + h = 0$. On the other side, if $W(a) + h = 0$, then $u(x, t)$ given by (4) satisfies (5). Therefore, it is an a – solution. $\square$

Theorem 1 provides the taxonomy of the equilibrium solutions, which is formulated as in Fig 4. This categorization was given as Theorem 2 in [1]. Its proof is based on graphical investigation of $W(x)$. We refer to [1, Appendix] for the proof.

³We used the change of variables $\tilde{y} = x - y$, $-d\tilde{y} = dy$ in the integral given in (3). This provides $\int_0^a \omega(x - y) dy = -\int_x^{x-a} \omega(\tilde{y}) d\tilde{y} = \int_{x-a}^x \omega(\tilde{y}) d\tilde{y} = \int_0^x \omega(\tilde{y}) d\tilde{y} - \int_0^{x-a} \omega(\tilde{y}) d\tilde{y} = W(x) - W(x - a)$.

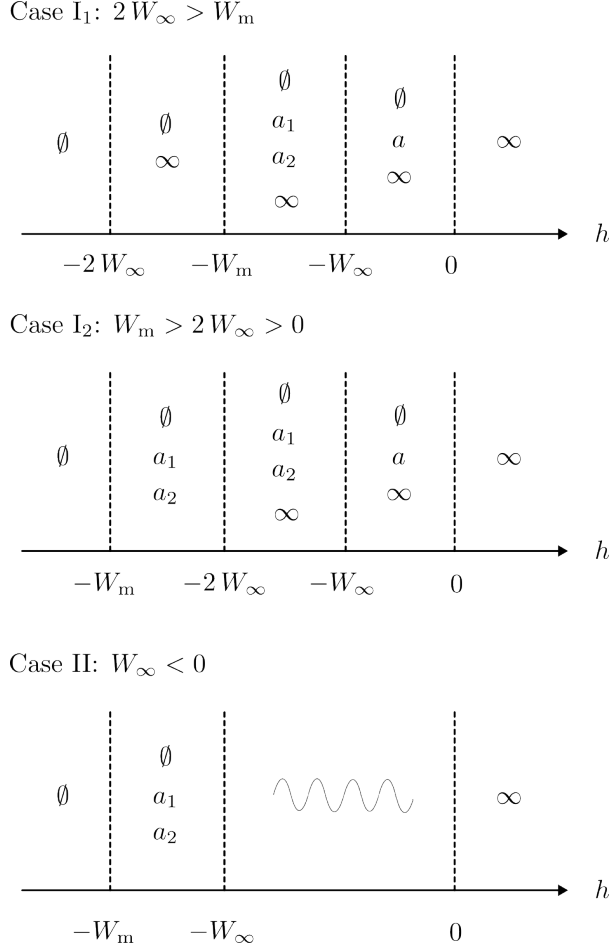


Figure 4: Taxonomy of equilibrium solutions for varying threshold stimulus h .

2 Stability of equilibrium solutions

Stability analysis focuses on how a small perturbation in initial condition of a system affects the solution. If a small perturbation in the initial condition does not introduce any change in the equilibrium solution, then the equilibrium solution is stable. Otherwise, it is unstable.

In the following, we will look at the behavior of a single peak of a solution $u(x, t)$, which is not necessarily stationary (not equilibrium). That is, it may be changing with t , so $\frac{\partial u(x, t)}{\partial t} = 0$ **need not** necessarily hold. Initial conditions (initial stimulation in Amari's terminology) will be spatially localized stimulation of the system (Fig 5).

Let the excited region at time $t \geq 0$ be

$$R_t(u(x, t)) = (x_1(t), x_2(t)),$$

We define the gradients at the boundaries of the excited region as follows (Fig 6):

$$c_1 := \frac{\partial u}{\partial x}(x_1, t), \quad c_2 := \frac{\partial u}{\partial x}(x_2, t). \quad (6)$$

At the end of a short time interval dt , the solution $u(x, t)$ evolves to $u(x, t + dt)$ (Fig 7). Therefore, the excited region evolves to

$$R_{t+dt}(u) = (x_1(t + dt), x_2(t + dt)).$$

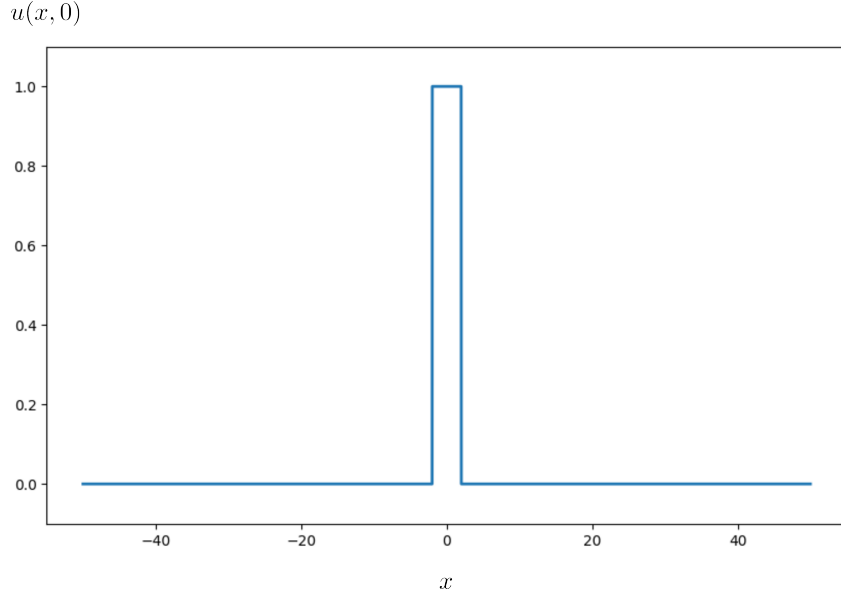


Figure 5: Plot of an example localized initial stimulation. This provides the initial condition for (2).

By definition, a localized excitation should be 0 at the boundaries x_1, x_2 ($x_2 > x_1$) of the excited region:

$$\begin{aligned} u(x_1(t), t) &= 0, & u(x_2(t), t) &= 0, \\ u(x_1(t + dt), t + dt) &= u(x_1 + dx_1, t + dt) = 0, \\ u(x_2(t + dt), t + dt) &= u(x_2 + dx_2, t + dt) = 0. \end{aligned} \quad (7)$$

Here dx_1, dx_2 and dt are infinitesimally small quantities. We approximate $u(x_1 + dx_1, t + dt)$ and $u(x_2 + dx_2, t + dt)$ by using their Taylor expansions around (x_1, t) and (x_2, t) , respectively:

$$\begin{aligned} u(x_1 + dx_1, t + dt) &\approx u(x_1, t) + dx_1 \frac{\partial u}{\partial x}(x_1, t) + dt \frac{\partial u}{\partial t}(x_1, t), \\ u(x_2 + dx_2, t + dt) &\approx u(x_2, t) + dx_2 \frac{\partial u}{\partial x}(x_2, t) + dt \frac{\partial u}{\partial t}(x_2, t), \end{aligned}$$

from which we can write the following due to (7):

$$\begin{aligned} dx_1 \frac{\partial u}{\partial x}(x_1, t) + dt \frac{\partial u}{\partial t}(x_1, t) &= 0, \\ dx_2 \frac{\partial u}{\partial x}(x_2, t) + dt \frac{\partial u}{\partial t}(x_2, t) &= 0. \end{aligned} \quad (8)$$

Finally, we use (7) in (1) ($P(x, t) = 0$ for all $x \in \mathbb{R}$ and $t \in [0, \infty)$ since we consider the cases in which there is no time-dependent external stimulation) to write

$$\begin{aligned} \mu \frac{\partial u}{\partial t}(x_1, t) &= \int_{x_1}^{x_2} \omega(x_1 - y) dy + h = W(x_2 - x_1) + h, \\ \mu \frac{\partial u}{\partial t}(x_2, t) &= \int_{x_1}^{x_2} \omega(x_2 - y) dy + h = W(x_2 - x_1) + h, \end{aligned} \quad (9)$$

where we used a change of variables similarly to (4), and $W(x) = -W(-x)$, to write the final equalities⁴.

⁴We use the change of variables $\tilde{y} = y - x_1$, which gives $d\tilde{y} = dy$. This provides $\int_{x_1}^{x_2} \omega(x_1 - y) dy = \int_0^{x_2 - x_1} \omega(-\tilde{y}) d\tilde{y} = \int_0^{x_2 - x_1} \omega(\tilde{y}) d\tilde{y}$, where we use the symmetry $\omega(x) = \omega(-x)$. This gives $\int_0^{x_2 - x_1} \omega(\tilde{y}) d\tilde{y} = W(x_2 - x_1)$. The same holds for the second line in (9).

From (8), we find:

$$\frac{dx_1}{dt} = -\frac{\frac{\partial u}{\partial t}(x_1, t)}{\frac{\partial u}{\partial x}(x_1, t)}, \quad \frac{dx_2}{dt} = -\frac{\frac{\partial u}{\partial t}(x_2, t)}{\frac{\partial u}{\partial x}(x_2, t)} \quad (10)$$

By using (9) and c_1, c_2 defined in (6), we write from (10) the following:

$$\begin{aligned} \frac{dx_1}{dt}(t) &= -\frac{\frac{\partial u}{\partial t}(x_1, t)}{\frac{\partial u}{\partial x}(x_1, t)} = -\frac{1}{\mu c_1} \left(W(x_2(t) - x_1(t)) + h \right), \\ \frac{dx_2}{dt}(t) &= -\frac{\frac{\partial u}{\partial t}(x_2, t)}{\frac{\partial u}{\partial x}(x_2, t)} = -\frac{1}{\mu c_2} \left(W(x_2(t) - x_1(t)) + h \right). \end{aligned} \quad (11)$$

Note that the length of the excited region at time t is (Fig)

$$a(t) = x_2(t) - x_1(t),$$

which implies

$$\frac{da}{dt}(t) = \frac{1}{\mu} \left(-\frac{1}{c_2} + \frac{1}{c_1} \right) \left(W(a(t)) + h \right), \quad (12)$$

where we used (11) and ignored the explicit time notation. This equation describes the dynamical description of the change of the length of the excited region. For an a -solution at equilibrium, the length a does not change in time, therefore $\frac{da}{dt} = 0$. This corresponds to an equilibrium length in (12). Since $c_1 > 0$ and $c_2 < 0$ (Figs 6 and 7) for an excited region, this happens only when

$$W(a) + h = 0,$$

which is in agreement with the third item of Theorem 1. Moreover, such equilibrium length a is **stable** if $\frac{dW(a)}{da} < 0$ and unstable if $\frac{dW(a)}{da} > 0$. This allows us to check the stability of equilibrium solutions easily by using the plot of $W(x)$ (Fig 3).

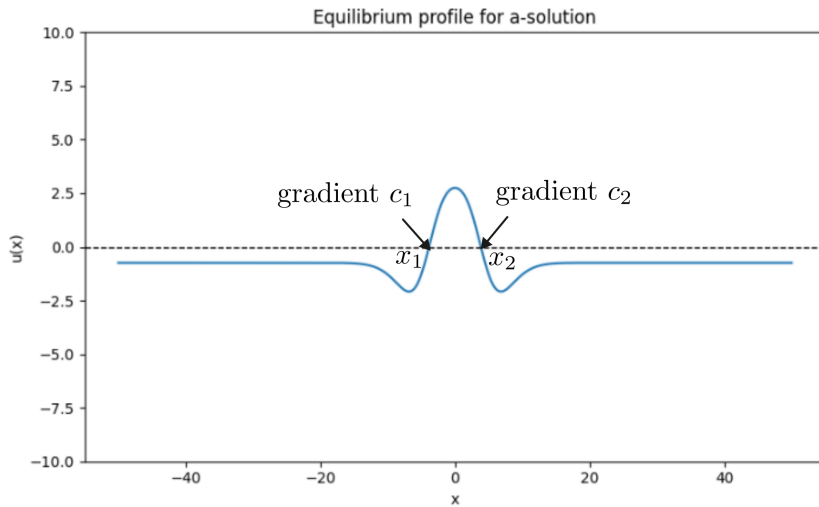


Figure 6: Gradients c_1 and c_2 at the boundaries of the localized excitation.

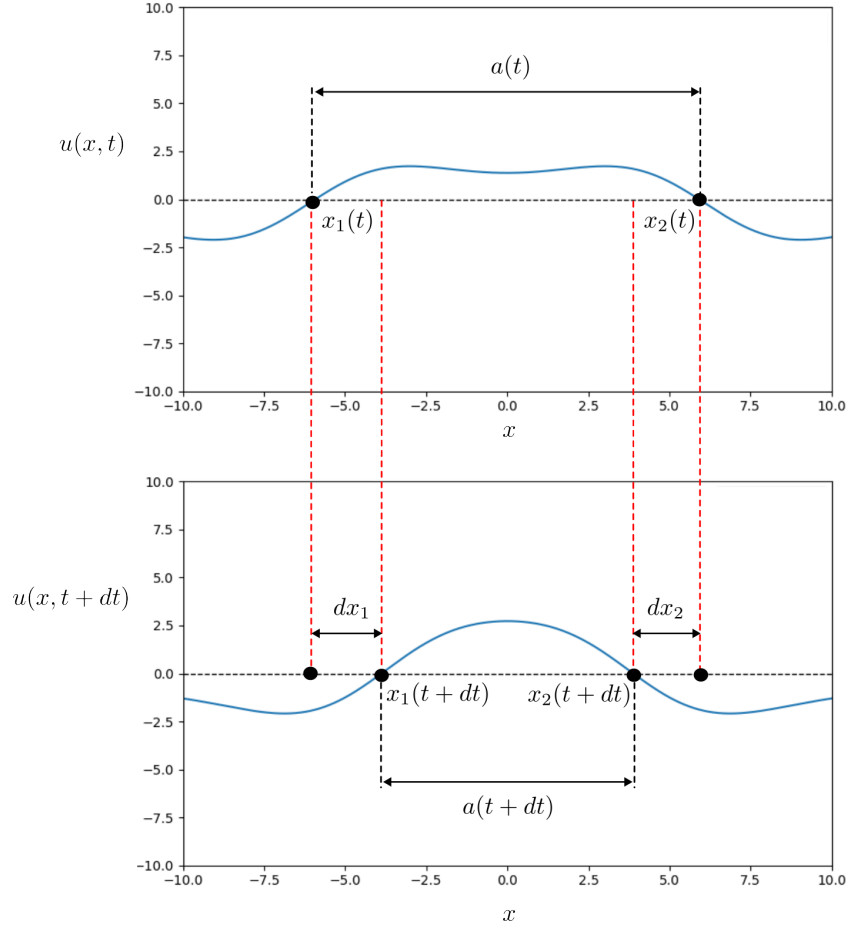


Figure 7: Evolution of a localized excitation during a short time interval dt before it reaches to equilibrium. Here $x_1(t), x_2(t)$ and $x_1(t + dt), x_2(t + dt)$ show the boundaries of the solution at time t and $t + dt$, respectively. The distances between the boundaries at time t and $t + dt$ are indicated by dx_1 and dx_2 . The length of the excited region at t and $t + dt$ is shown by $a(t)$ and $a(t + dt)$, respectively.

Example 1.

We can think of a case where h is chosen such that there are two a values a_1 and a_2 ($a_2 > a_1$) for an a – solution:

$$W(a_1) + h = 0, \quad W(a_2) + h = 0.$$

Inspection of the stability of these solution can be done directly on the corresponding $W(x)$ graph. In Fig 8, we see that the gradient for W at a_1 is positive ($\frac{dW}{da}(a_1) > 0$). Therefore, a_1 is an unstable equilibrium of (12), i.e., a_1 – solution is an unstable equilibrium solution. This is the opposite for a_2 , therefore a_2 is a stable equilibrium of (12), thus a_2 – solution is a stable equilibrium solution. $\diamond$

For a finite excited region to explode towards an ∞ – solution, the following condition is necessary:

$$W_\infty + h > 0. \tag{13}$$

If this condition is not satisfied, an ∞ – solution cannot be obtained without stimulating infinitely large regions.

Therefore, we ignore ∞ – solution in the cases where (13) is not satisfied. This simplifies the taxonomy given in Fig 4, removing the elements with $h < -W_\infty$ from Cases I₁ and

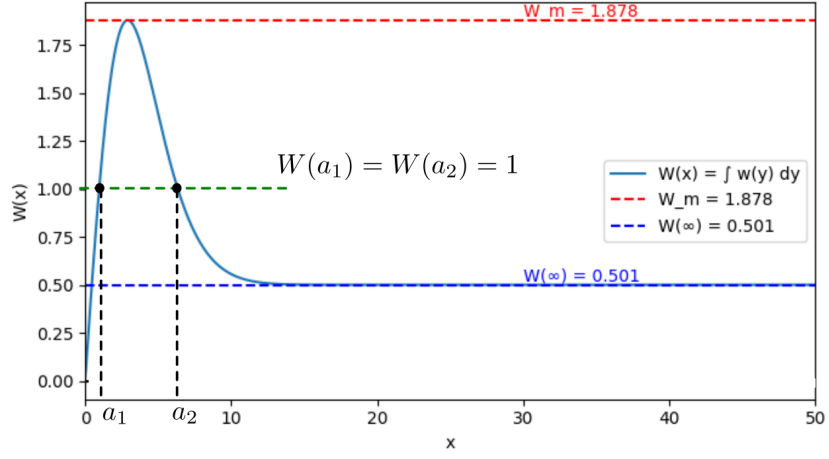
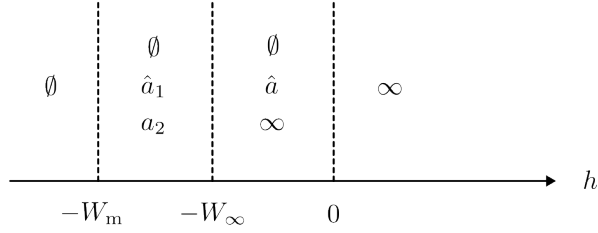


Figure 8: Plot of $W(x)$ where h is chosen ($= -1$) such that $W(a_1) + h = W(a_2) + h = 0$. This plot provides a graphical explanation of Example 1.

I_2 . So the Cases I_1 and I_2 overlap. We provide in Fig 9 this simplified version in which we denote the unstable equilibrium solutions with the hat sign $\hat{}$.

Case I: $W_\infty > 0$



Case II: $W_\infty < 0$

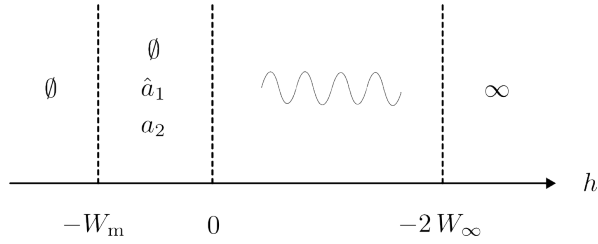


Figure 9: Simplified taxonomy of equilibrium solutions. The hat sign $\hat{}$ indicates the unstable solutions.

References

- [1] S.-i. Amari, “Dynamics of pattern formation in lateral-inhibition type neural fields,” *Biological Cybernetics*, vol. 27, no. 2, pp. 77–87, 1977.